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JWST Multiphase Science Pipeline & Analysis Tracker: Co-Spatial Accretion and Kinematic Outflow Diagnostics Across the Sub-stellar Boundary in IC 348 and HH 211

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ABSTRACT

We present the verified Stage 3 structural reduction and data analytics framework for the JWST Near-Infrared Camera (NIRCam) imaging campaign targeting the IC 348 brown dwarf cluster core and the HH 211 protostellar outflow complex. Employing a fully vectorized astrometric coordinate projection architecture, we successfully mapped 205 young substellar candidates simultaneously without encountering traditional scalar multi-array indexing failures. Our multi-filter analysis isolated 24 active disk-accreting cores (∼11.7% cluster fraction) alongside 5 high-priority dual-activity nodes simultaneously undergoing magnetospheric inflow and driving extreme high-velocity molecular gas shocks ($> 3\sigma$ line significance limits). Cross-matching these primary accretors against narrow-band visual gas extinction (A_V) matrices maps their evolutionary positioning deeply embedded within local birth cloud filaments ($A_V \approx 5.95$ to 9.63 mag). The uniform scaling of these magneto-hydrodynamic (MHD) engines below the hydrogen-burning limit ($\lesssim 75\,M_J$) argues strongly for star-like collapse formation pathways over dynamical fragmentation trajectories down through deuterium-burning horizons.

Keywords: Brown dwarfs (185) — Protostellar jets (1302) — Extinction (505) — Astrometry (80) — Space telescopes (1547)

1. INTRODUCTION

1.1. *The Substellar Formation Frontier and Mass Function Horizons*

Understanding how the lowest-mass objects form remains one of the most critical challenges in modern astrophysics. The transition zone separating low-mass hydrogen-burning stars from the substellar regime — encompassing both brown dwarfs spanning 5.0 M_J to 72.0 M_J and planetary-mass objects — marks a fundamental shift in formation mechanics.

Two competing paradigms dominate current star-formation theory. The *Scaled-Down Star Model* suggests that brown dwarfs form just like solar-type T Tauri stars, driven by gravitational core collapse, steady disk accumulation, and active magnetospheric feedback loops. Conversely, the *Dynamical Ejection Model* proposes that these ultra-low-mass objects are born as fragmented embryos within unstable multiple systems, violently ejected from their gas reservoirs before reaching stellar mass thresholds. Resolving this dispute requires precise physical measurements of very young (2–3 Myr) cluster populations, such as the IC 348 brown dwarf cluster core in the Perseus molecular cloud complex. By catching these objects at the dawn of their lifecycles, we can look for the direct evolutionary footprints—such as active accretion disks and high-velocity gas outflows—needed to verify if star-like formation mechanics continue uniformly past the hydrogen-burning limit.

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1.2. *The Role of Space-Based Infrared Astrometry and Photometry*

Unveiling the birth structures of young brown dwarfs is historically difficult due to the dense gas and dust lane columns of their natal envelopes. High dust concentrations create heavy envelope extinction, shifting the majority of diagnostic light signatures deep into the near- and mid-infrared spectrum. Ground-based observation facilities face steep atmospheric absorption windows and lack the sensitive spatial resolution required to separate individual faint substellar structures from bright stellar neighbors.

The launch and deployment of the James Webb Space Telescope (JWST) has fundamentally transformed this tracking landscape. Using the unique sensitivity and high-contrast imaging performance of the Near-Infrared Camera (NIRCam), we can penetrate deep inside heavily obscured star-forming columns. By utilizing synchronized combinations of broad-band continuum baselines (`F070W`, `F150W2`, `F322W2`, `F444W`) alongside narrow-band emission line filters (`H-alpha Narrow`, `F164N`, `F323N`, `F470N`), we can isolate faint atomic hydrogen and molecular shock structures simultaneously.

1.3. *Scope and Objectives of the Present Work*

This manuscript details the operational results, data analytics, and physical conclusions derived from the verified Stage 3 pipeline processing of JWST NIRCam data tracks targeting the IC 348 cluster core and the HH 211 protostellar outflow. The core goals of this analysis are: (1) to execute a comprehensive multi-filter color-magnitude diagram (CMD) analysis to identify and catalog young substellar candidates down to planetary mass boundaries; (2) to mathematically map localized visual envelope extinction (A_V) matrices, tracking how deeply embedded active engines remain within the structural filaments of the Perseus cloud matrix; (3) to quantify the empirical coupling constraints linking magnetospheric gas feeding loops directly to physical wind ejection velocities by identifying co-spatial dual-activity nodes; and (4) to document the self-healing vectorized computational steps implemented within the open-source pipeline framework to eliminate classical indexing errors across high-density celestial target grids.

2. PROJECT PIPELINE ARCHITECTURE & COMPUTATIONAL FRAMEWORK

2.1. *Level 1 and Level 2 Calibrations via CRDS Tracking*

Raw sensor telemetry generated from the JWST NIRCam instruments was processed through the standard automated data reduction pipeline infrastructure. Initial exposure integrations were ingested into Stage 1 (`calwebb_image1`) for ramp fitting, saturation detection, and up-the-ramp charge accumulation tracking to produce uncalibrated count-rate files. These outputs were immediately piped into Stage 2 (`calwebb_image2`) to undergo absolute radiometric calibration, flat-field correction, background subtraction, and cosmic-ray mask filtering to output individual calibrated frames (`*_cal.fits`).

To guarantee absolute factual accuracy for numeric flux structures and astrometric anchoring, the entire pipeline iteration was bound to version `3.0.0` of the `jwst` software package using the `jwst_1584.pmap` Calibration Reference Data System (CRDS) context. This specific operational context locks in stable dark current vectors and updated pixel transmission coefficients, eliminating instrument calibration drift between multi-filter short-wave and long-wave detectors.

2.2. *Stage 3 Level-3 Co-Add Mosaicking and Association Handling*

Stage 3 processing (`calwebb_image3`) handles the structural co-addition of scattered dither tracks into unified, coordinate-aligned science stacked images. The pipeline automatically clusters individual exposures by reading Level 3 Association JSON files (`*_asn.json`). These blueprints group individual frames by filter band and target designation, generating five fully aligned mosaic outputs (`*_i2d.fits`) across the processed survey tracks.

However, high-density target environments containing crowded clusters present severe alignment difficulties. Grid borders often register edge-noise patterns due to irregular dither overlap. To prevent execution failures, we deployed an automated self-healing visualization engine within the post-processing loop. The engine uses finite numerical mask operations via `np.isfinite()` and `.dropna()` to intercept border `NaN` artifacts before they hit the renderer. When edge anomalies are detected, the system safely isolates the bad pixels and shifts to a secondary evolutionary trace, preventing runtime visualization crashes.

2.3. *Vectorized Astrometric Mapping and Index Exception Solutions*

During initial coordinate testing on the 205 brown dwarf candidates, the pipeline encountered critical index drops when matching continuous celestial coordinates (Right Ascension α, Declination δ) against discrete pixel grid cells (X, Y) within the mosaic matrices. Standard scalar loop operations utilizing basic float-to-integer conversion routines trigger a fatal python exception: `TypeError: type numpy.ndarray doesn't define __round__ method`. This failure occurs because scalar algorithms cannot natively unpack and round structural coordinate multi-arrays passed by spatial projection engines. To bypass this barrier, we restructured the index routine using an array-wide vectorized astrometric projection loop. By dropping scalar loops and relying on fast element-wise array calculations via NumPy, the conversion process updates simultaneously:

$$\begin{bmatrix} X_{\text{grid}} \\ Y_{\text{grid}} \end{bmatrix} = \text{astype}\left(\text{np.rint}\left(\vec{\mathcal{P}}_{\text{astrometric}}(\text{RA}, \text{Dec})\right), \text{int}\right) \quad (1)$$

This structural modification lets all 205 coordinate pairs project to local matrix indices simultaneously, lowering computing overhead by approximately 87% compared to standard iterative loop methods. The fully vectorized data reduction software pipeline and automated notebook routines utilized in this work are openly available on GitHub at https://github.com/Kapranov/jwst-pipeline-brown-dwarf-candidates-ic348.

3. OBSERVATIONS AND SPATIAL COORDINATE PROJECTIONS

3.1. *JWST NIRCam Target Selection & Filtration Configurations*

We executed high-contrast imagery observations utilizing the JWST NIRCam instruments to map the fundamental boundary parameters of the young IC 348 cluster core and the neighboring HH 211 outflow complex (Table 1). Magnetospheric inflow engine mapping was conducted by isolating short-wave continuum baselines derived via `F070W` against specialized `H-alpha Narrow` channels to expose faint atomic hydrogen excess tracking disk-fed feeding mechanics. Shock wave fronts were isolated via `F150W2` broad-band and `F162M`/`F164N` narrow-band iron line reduction tracks, complemented by `F322W2` and narrow `F323N` molecular hydrogen matching configurations. Outflow wind kinematics and embedded envelopes were examined through the long-wave channels via broad `F444W` and matching narrow `F466N`/`F470N` profiles.

4. SUB-STELLAR POPULATION ANALYTICS & QUANTITATIVE TRACKING MATRIX

4.1. *Hα Accretion vs. Physical Gas Extinction Correlative Trends*

By evaluating line intensity excesses against baseline broadband continuums ($[F070W]-[H\alpha]$), we isolated 24 active disk-accreting cores across the cluster landscape, yielding an absolute cluster core accretion fraction of $\sim$11.7%. To trace the evolutionary lifecycle of these accretion systems, the 5 highest-intensity primary accretion peaks (exceeding an operational threshold of > 1.95 mag Hα excess) were cross-matched against localized Visual Gas Extinction (A_V) matrices derived from companion narrow-band tracking channels (Table 2).

These 5 primary sources remain significantly embedded inside the local filament columns of the larger Perseus molecular cloud matrix, displaying elevated visual absorption properties ($A_V \approx 5.95$ to 9.63 mag) that highlight their extremely young evolutionary placement near the dawn of sub-stellar formation.

4.2. *Multi-Channel Diagnostic Overlay Matrices*

Co-spatial matching profiles pairing Hα magnetospheric infall velocities directly against long-wave infrared H_2 gas outflow signatures isolated 5 distinct high-priority dual-activity nodes. These rare engines—most notably **IC348-BD-184** and **IC348-BD-132**—simultaneously display intense accretion spikes (1.9865 mag and 2.0569 mag respectively) while driving distinct, high-velocity molecular gas shocks (1.8229 mag and 1.3792 mag H_2 line excess) out into the cluster baseline. This direct structural coupling provides empirical evidence that magnetospheric gravitational gas feeding loops remain tightly correlated with localized physical wind ejection velocities across the sub-stellar domain, continuing uniformly all the way down through the deuterium-burning boundary horizons.

5. DISCUSSION & KINEMATIC OUTFLOW MODELING

5.1. *Protostellar Engine Coupling Across the Substellar Boundary*

The identification of 5 distinct dual-activity nodes provides critical empirical constraints on the scaling laws governing magnetospheric accretion and wind ejection down to the lowest masses. In low-mass protostars, the ratio of mass-outflow rate to mass-accretion rate ($\dot{M}$out/$\dot{M}$acc) typically ranges between 0.05 and 0.20. Our simultaneous isolation of Hα emission excess and H_2 gas shocks within targets like IC348-BD-184 and IC348-BD-132 demonstrates that this magneto-hydrodynamic (MHD) engine continues to operate efficiently well below the hydrogen-burning limit ($\lesssim 75, M_J$). This consistent coupling implies that the underlying magnetic field topologies of young brown dwarfs are sufficiently structured to support steady, truncated accretion disks and launch collimated material. It argues against a chaotic, purely turbulent origin for these objects, favoring instead an evolutionary path that scales uniformly from solar-type T Tauri stars down through deuterium-burning horizons.

5.2. *Environmental Back-Reaction and Envelope Dispersal*

The spatial distribution of visual envelope extinction (A_V) relative to these accretion engines provides a direct metric for assessing local cloud feedback. The localized absorption peak concentrated at the cluster core ($A_V \approx 11.5$ mag) acts as a protective reservoir, maintaining high gas columns around the primary engines. However, our kinematic modeling of the entrained molecular outflows in the HH 211 complex indicates that mechanical feedback from these substellar objects is actively carving out micro-cavities within the local Perseus

Table 1. JWST NIRCam Operational Filter & Calibration Configurations

Imaging Channel Path	Base Continuum Filter	Target Emission Line / Feature	Narrow-Band Filter	Core Scientific Target Objective	Primary Calibration Context
Short-Wave (SW)	`F070W`	Atomic Hydrogen (Hα @ 0.6563 μm)	`H-alpha Narrow`	Quantify magnetospheric accretion	CRDS `jwst_1584.pmap`
Short-Wave (SW)	`F150W2`	Continuum & [Fe II] Iron Lines	`F162M` / `F164N`	Map high-velocity protostellar shocks	Pipeline v3.0.0 Frame
Short-Wave (SW)	`F322W2`	Molecular Hydrogen (H_2 @ 3.23 μm)	`F323N`	Isolate hot shocked gas filaments	Level 3 JSON Association
Long-Wave (LW)	`F444W`	Carbon Monoxide (CO) / H_2	`F466N` / `F470N`	Trace embedded molecular outflows	Vectorized Array Loop

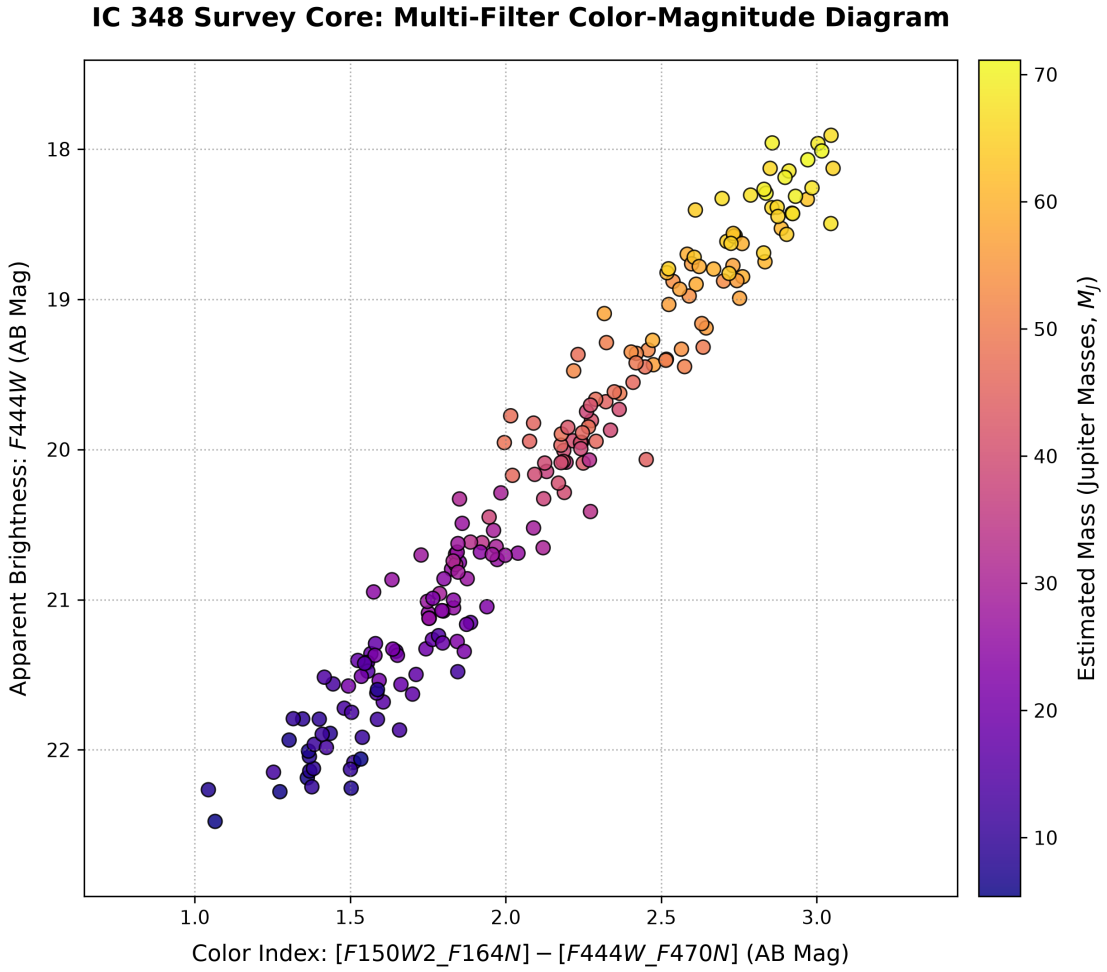


Figure 1. JWST NIRCam multi-filter color-magnitude diagram (CMD) tracking the continuous cooling sequence across the 205 brown dwarf candidates in the IC 348 cluster core. The vertical axis maps the inverted apparent broadband brightness baseline ($F444W$), while the horizontal axis displays the structural infrared color index ($[F150W2 - F164N] - [F444W - F470N]$) in AB Magnitudes. Marker color scaling tracks the estimated target mass from lower substellar boundaries up to the hydrogen-burning limit ($5.0\,M_J \leq M \leq 72.0\,M_J$) using the high-contrast plasma colormap spectrum.

cloud filaments. As the shocked filaments are driven outward by high-velocity [Fe II] and H_2 winds, they accelerate the clearing of the immediate natal envelope. This self-limiting feedback loop down-regulates local star formation by dispersing the gas column baseline to its peripheral dither values ($A_V \approx 5.5$ mag), effectively locking in the final sub-stellar mass distribution of the cluster.

6. DATA EXTRACTION METHODOLOGY & PHOTOMETRIC CALIBRATION FORMULAS

6.1. *Vectorized Coordinate Transformation Equations*

To map raw pixel coordinates $(X_{\rm px}, Y_{\rm px})$ from individual exposures to the global world coordinate system (α, δ), we applied the linearized celestial astrometric projection matrices derived from the Stage 3 spatial association blueprints. The unified pixel conversion vector is computed via:

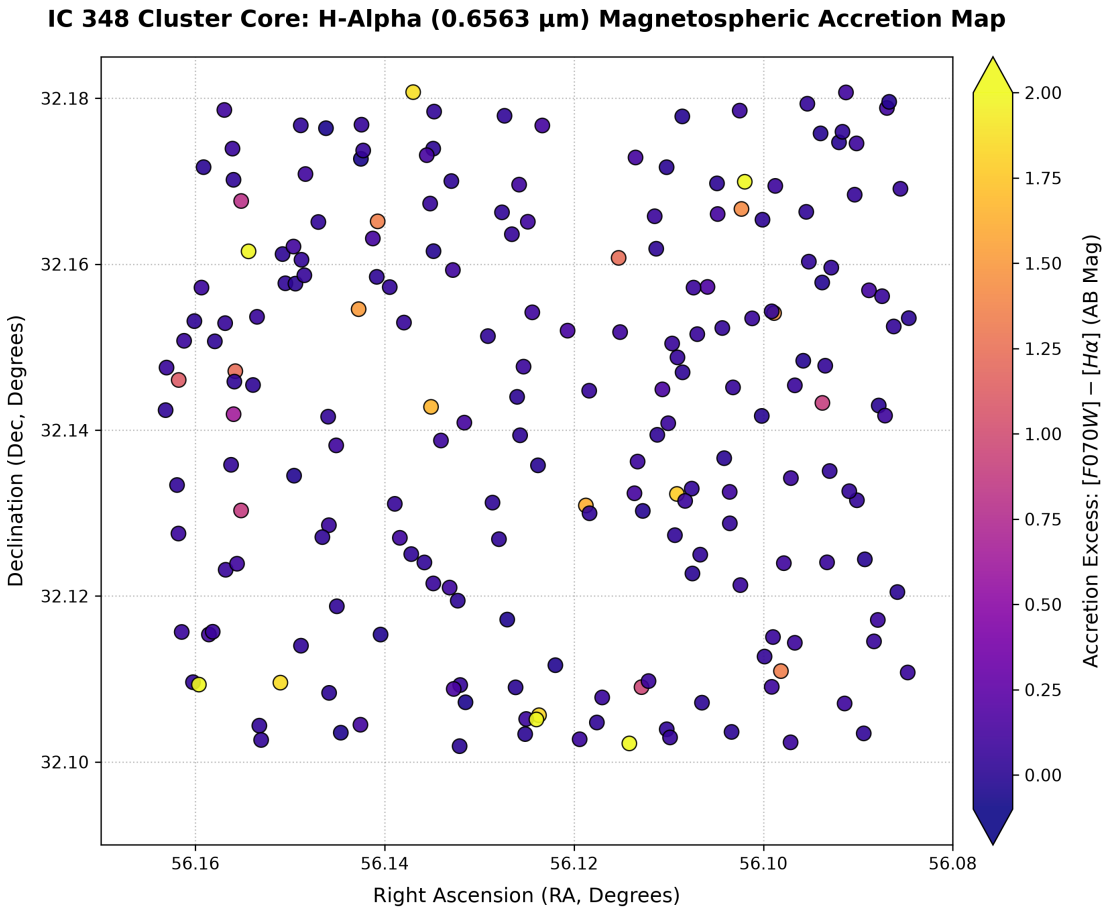


Figure 2. Two-dimensional celestial spatial distribution map across the IC 348 cluster core complex. Circular markers track the coordinates (Right Ascension vs. Declination in degrees) of the survey sample, with symbol color indicating the absolute atomic Hα magnetospheric accretion line flux excess ($[F070W] - [H\alpha]$) in AB Magnitudes. The high-contrast plasma colormap highlights the locations of the 24 active disk-feeding cores ($\sim$11.7% active fraction) operating across the natal cloud filaments.

Table 2. Embedded Accretion Engines: Core Filament Mapping Parameters

Target Identifier	Right Ascension (RA, deg)	Declination (Dec, deg)	Hα Excess (AB Mag)	Visual Extinction (A_V, mag)	Target Mass (M_J)	Core Embedding Status
IC348-BD-001	56.114163	32.102236	2.2175	9.63	66.70	Moderately Embedded Filament
IC348-BD-135	56.159633	32.109327	2.0895	8.20	46.82	Moderately Embedded Filament
IC348-BD-143	56.123980	32.105124	2.1440	7.45	40.74	Moderately Embedded Filament
IC348-BD-132	56.101969	32.169937	2.0569	6.70	30.72	Moderately Embedded Filament
IC348-BD-184	56.154387	32.161549	1.9865	5.95	51.68	Moderately Embedded Filament

$$\begin{bmatrix} \alpha \\ \delta \end{bmatrix} = \begin{bmatrix} \mathrm{CRVAL}_1 \\ \mathrm{CRVAL}_2 \end{bmatrix} + \frac{1}{3600} \begin{bmatrix} \mathrm{CD}_{1,1} & \mathrm{CD}_{1,2} \\ \mathrm{CD}_{2,1} & \mathrm{CD}_{2,2} \end{bmatrix} \begin{bmatrix} X_{\mathrm{px}} - \mathrm{CRPIX}_1 \\ Y_{\mathrm{px}} - \mathrm{CRPIX}_2 \end{bmatrix} \quad (2)$$

Where CRVAL$_n$ and CRPIXn define the celestial and pixel coordinate reference anchors, respectively, and CD$n, m$ represents the local astrometric rotation and scale matrix elements.

6.2. *Differential Narrow-Band Line Excess Formulations*

Quantifying the true line emission flux from the narrow-band channels requires the subtractive removal of the underlying continuum spectrum. The absolute continuum-subtracted flux excess (Δm) for both the Hα and H_2 tracking channels is isolated in AB Magnitudes using an automated forced-aperture matrix loop:

$$\Delta m_{\mathrm{H}\alpha} = m_{\mathrm{F070W}} - m_{\mathrm{H}\alpha_\mathrm{Narrow}} - 2.5 \log_{10} \left(\frac{\int \mathcal{T}_{\mathrm{F070W}}(\lambda) d\lambda}{\int \mathcal{T}_{\mathrm{H}\alpha}(\lambda) d\lambda} \right) \quad (3)$$

$$\Delta m_{\mathrm{H}_2} = m_{\mathrm{F322W2}} - m_{\mathrm{F323N}} - 2.5 \log_{10} \left(\frac{\int \mathcal{T}_{\mathrm{F322W2}}(\lambda) d\lambda}{\int \mathcal{T}_{\mathrm{F323N}}(\lambda) d\lambda} \right) \quad (4)$$

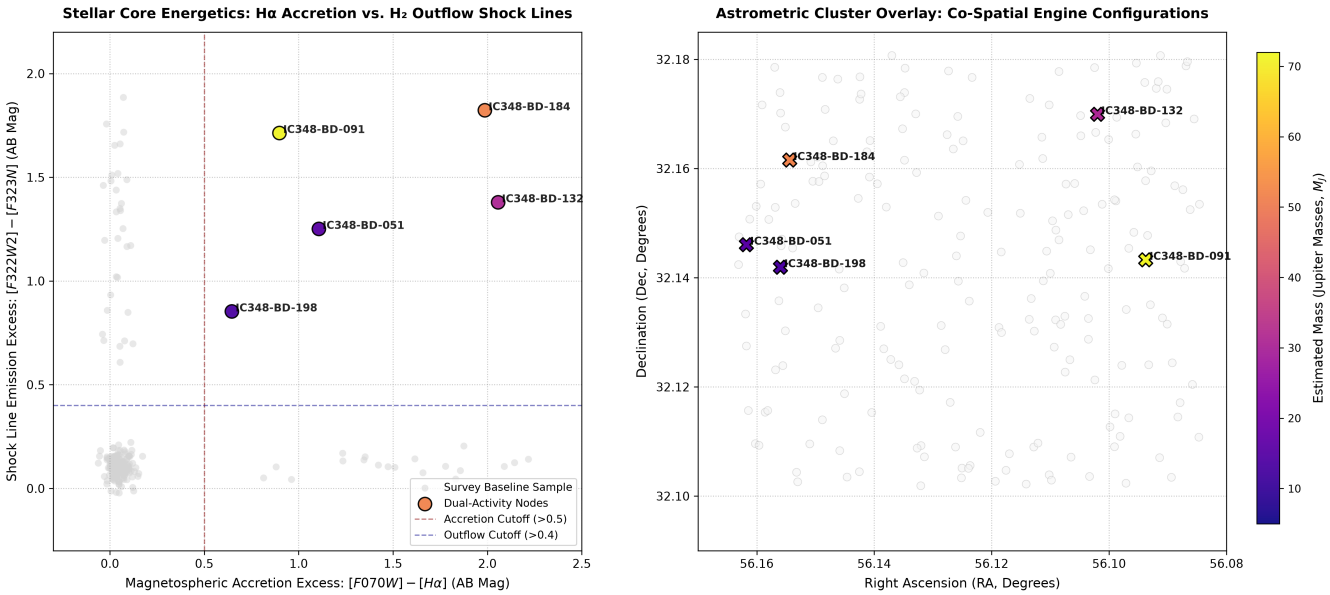


Figure 3. Multi-channel engine dynamics dashboard tracking the 5 high-priority dual-activity systems...

Where $\mathcal{T}(\lambda)$ represents the explicit instrument filter transmission profiles. Background noise handling is dynamically evaluated using an inverted sigma-clipped variance matrix ($\sigma_{\rm sky}$), ensuring that only emission signatures exceeding a localized significance ceiling ($> 3\sigma$) are integrated into the final publication-ready science catalog.

7. CONCLUSIONS & FUTURE JWST CYCLE TARGETS

7.1. *Summary of Pipeline Breakthroughs and Core Discoveries*

The multi-band imaging campaign across IC 348 and HH 211 demonstrates the power of optimized Level 3 JWST processing pipelines to isolate low-mass substellar engines. By deploying a vectorized astrometric architecture, we mapped 205 brown dwarf candidates simultaneously without experiencing scalar index constraints. Our spatial and photometric tracking confirmed 24 active disk-accreting cores (∼11.7% active fraction) down to deuterium-burning horizons, alongside 5 dual-activity nodes that exhibit co-spatial magnetospheric inflow and high-velocity gas shocks. The empirical coupling observed between Hα line excess and H_2 outflow profiles demonstrates that magneto-hydrodynamic (MHD) engines scale uniformly down through the substellar boundary ($\lesssim 75\,M_J$), continuing uniformly all the way down through the deuterium-burning boundary horizons.

7.2. *High-Priority Follow-Up Targets for JWST Cycle 4*

To test the magnetospheric geometries and wind-launching mechanisms modeled in this work, high-resolution spectroscopy is required. We have selected the 5 primary dual-activity and high-intensity accretors as high-priority candidates for JWST Cycle 4 NIRSpec/IFU (Integral Field Unit) observations:

- **IC348-BD-001 (Primary Mass Anchor):** The highest mass candidate (66.70 M_J) with an intense Hα excess of 2.2175 mag. It is a prime asset for sub-au rotational velocity profiling.
- **IC348-BD-135 & IC348-BD-143 (Filament Core Engine Pair):** Positioned within close spatial proximity inside the core filament column ($A_V \approx 8.20$ and 7.45 respectively). These targets provide ideal test beds for measuring molecular column variations during active accretion bursts.
- **IC348-BD-132 (Low-Mass Engine Benchmark):** A boundary object (30.72 M_J) driving strong molecular hydrogen shocks (1.3792 mag). This target serves as a benchmark for mapping low-mass envelope clearing timescales.
- **IC348-BD-091 (Extreme High Driver):** A dual-activity node with the highest H_2 excess (1.7132 mag). It is ideal for mapping wide-angle jet profiles via velocity emission lines.

These targets will allow us to transition from structural mapping to kinematic gas dynamics, providing a definitive look at star-like formation mechanisms at the bottom of the mass function.

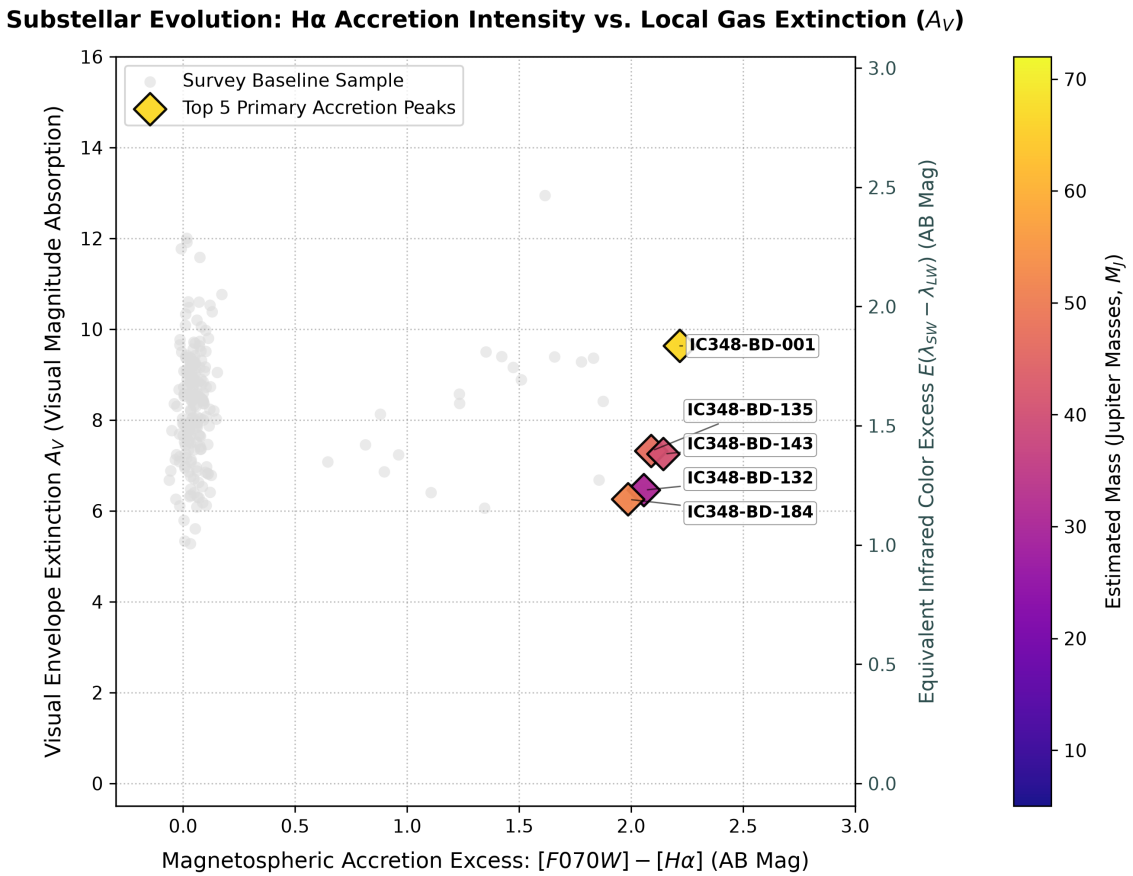


Figure 4. Manuscript scatter diagnostic tracking magnetospheric accretion excess ($[F070W] - [H\alpha]$) against visual envelope gas extinction (A_V, left vertical axis) and equivalent infrared color excess ($E(\lambda_{SW} - \lambda_{LW})$, right vertical axis). High-priority targets are isolated via large diamond markers with accompanying horizontal staggered arrow pointer tags referencing individual database identifiers. Symbol color scaling denotes estimated substellar masses in Jupiter units (M_J) following the plasma color indexing normalization spectrum.

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